

INDIAN INSTITUTE OF TECHNOLOGY BOMBAY

Department of Energy Science and Engineering

SELF-INITIATED PROJECT

Energy Stock Price Forecasting Using LSTM Neural Networks

Submitted by	Aarya Parekh
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Abstract

This project applies Long Short-Term Memory (LSTM) neural networks to forecast the next-day closing price of NTPC — India's largest thermal power utility and the highest-weight constituent of the NIFTY Energy Index. Five years of daily OHLCV data are sourced via the yfinance API and enriched with 17 technical indicators spanning trend, momentum, volatility, and volume categories. The enriched feature set is split chronologically (70/15/15) to prevent data leakage, with StandardScaler normalization fitted on the training set alone.

Two baseline models — Linear Regression (RMSE 5.44) and Random Forest (RMSE 24.85) — are trained first to establish reference performance. A 2-layer LSTM with 64 hidden units, 30-day lookback, and early stopping achieves RMSE 18.16 on the held-out test set. The primary finding is that Linear Regression outperforms both non-linear models because the dominant features (SMA_50, SMA_200, EMA_12) are smoothed proxies of the target variable, making the underlying mapping near-linear.

The project demonstrates proficiency in data engineering (yfinance, pandas), machine learning (scikit-learn), deep learning (PyTorch), statistical analysis (ADF test, ACF/PACF), and version control (Git/GitHub). Future work includes return-based prediction, multi-stock forecasting, and backtesting with Sharpe ratio evaluation.

Keywords: LSTM, stock price forecasting, time-series analysis, technical indicators, NTPC, energy sector, PyTorch, machine learning.

1. Introduction and Motivation

The Indian energy sector has seen sustained growth over the past decade, driven by increasing electrification, industrial expansion, and policy initiatives. NTPC, as the country's largest power generator with an installed capacity exceeding 70 GW, serves as a bellwether for the sector. Accurately forecasting its stock price has implications both for portfolio construction and for understanding market dynamics in regulated-utility equities.

Classical econometric approaches (ARIMA, GARCH) assume stationarity and linear dependence. While effective for short-horizon volatility modelling, they struggle to capture the non-linear, regime-switching behaviour common in equity markets. Deep learning models — particularly LSTMs — have emerged as a flexible alternative. Their gating mechanism allows selective retention of information across long sequences, making them well-suited for time-series forecasting tasks where the relevant lookback window is unknown a priori.

This project builds a complete machine-learning pipeline from scratch: data acquisition, exploratory analysis, feature engineering, baseline benchmarking, LSTM training, and performance evaluation. The objective is twofold: (i) to produce a working forecasting system, and (ii) to develop a thorough understanding of each component's role and limitations.

2. Dataset

Item	Detail
Stock	NTPC.NS (National Stock Exchange)
Period	5 years of daily data (~1,250 trading days)
Fields	Open, High, Low, Close, Volume (OHLCV)
Source	Yahoo Finance via yfinance Python API
Proxy rationale	Highest-weight stock in NIFTY Energy Index

3. Methodology

3.1 Exploratory Data Analysis

Six visualisations are produced: closing price trend, daily returns, 30-day rolling annualised volatility, returns distribution (histogram + KDE), monthly returns heatmap, and ACF/PACF autocorrelation plots. An Augmented Dickey-Fuller test confirms that raw prices are non-stationary ($p > 0.05$) while daily returns are stationary ($p < 0.05$).

3.2 Feature Engineering

17 technical indicators computed using the **ta** library:

Category	Indicators
Trend (6)	SMA(20, 50, 200), EMA(12, 26), MACD
Momentum (2)	RSI(14), Stochastic K
Volatility (3)	Bollinger Bands (upper, lower), ATR
Volume (1)	On-Balance Volume (OBV)
Custom (5)	Returns(1d, 5d, 20d), Volume ratio, Price range

3.3 Data Splitting and Scaling

Chronological split — 70% train, 15% validation, 15% test — no shuffling. Separate StandardScalers for features and target, both fit on training set only.

3.4 Baseline Models

Linear Regression as simplest baseline. **Random Forest** (200 trees, max_depth=10) as non-linear ensemble. Both evaluated on RMSE, MAE, directional accuracy.

3.5 LSTM Model

2-layer LSTM, 64 hidden units, 30-day sliding window. Dense output (64→32→1). Adam (LR=0.001), ReduceLROnPlateau (patience=5), gradient clipping (max_norm=1.0), early stopping (patience=10).

4. Results

Model	RMSE	MAE	Dir. Accuracy
Linear Regression	5.44	4.06	51.74%
Random Forest	24.85	22.30	49.57%
LSTM (2-layer, 64h)	18.16	14.08	50.00%

Linear Regression achieves the lowest RMSE because the dominant features (SMA_50, SMA_200, EMA_12 — 77% of feature importance) are smoothed versions of the closing price itself, making the mapping near-linear.

5. Project Structure

Section	Title	Status
1	Data collection (downloaddata.py)	Complete
2	EDA (exploredata.py, returns_analysis.py, eda_remaining.py)	Complete
3	Feature engineering (feature_engineering.py)	Complete
4	Baseline models (baseline_models.py)	Complete
5	LSTM model (lstm_model.py)	Complete
6	Backtesting and trading strategy	Planned
7	README and documentation	Complete

6. Reference List

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Aarya Parekh

B.Tech Energy Engineering, Roll No. 24B1535

Date: _____